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An in vitro approach for simulating divergent Golgi *O*-glycosylation of tumor-associated MUC1 from normal MUC1

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Abdullateef Nashed[1,2], Kyllen Dilsook[1,2], Tharindu Senapathi[1,3] & Kevin J. Naidoo[1,2]

Peptide *O*-glycosylation relies on the coordinated action of glycosyltransferases across the endoplasmic reticulum (ER) and Golgi apparatus. However, the molecular mechanisms driving aberrant glycosylation in cancer remain poorly understood. Here we show an in vitro one-pot synthetic biology approach that simulates divergent glycosylation pathways to map the synthesis of mucin 1 (MUC1) tumor-associated antigens. By modeling the cancer-associated relocation of initiation enzymes (GALNTs) to the ER, we demonstrate that this spatial shift leads to complete GalNAc (Tn antigen) occupancy. This occurs because ER localization extends reaction times and prevents inhibition by downstream Golgi enzymes. Furthermore, combined kinetic and computer reaction dynamic simulations reveal that ST6GALNAC1 exclusively drives α−2-6 sialylation, with a strict preference for the T13 site on fully glycosylated MUC1. This suggests that cancer-associated sTn upregulation is directly linked to T13 occupancy. Ultimately, this systems modeling approach decodes the enzyme localization and substrate specificities fundamental to tumourigenesis.

The post-translational event of peptide or protein glycosylation is a non-template-driven process that relies on more than just glycoenzyme gene expression data or the actual glycoenzyme expression levels themselves[1]. A case in point is that while glycosyltransferase (GT) differential gene expressions can be used to classify cancer[2], this genomic-level data cannot be used to directly infer the difference between cancerous and healthy glycoconjugate expression. Specifically, the characteristically high degree of sialylation observed in tumor tissues[3] and the associated structural modifications of glycans cannot be directly correlated with the genes that express the sialyltransferases. This is because the complex glycosylation pathways within cells are intimately connected and intertwined with their critical metabolic and regulatory networks[4]. This complexity dissociates the high degree of sialylation observed in tumor tissues[3] and structural modifications of glycans from the direct gene and protein expression readout. Consequently, a critical first step in systems biology modeling of glycosylation is the construction of a developmental model to mimic sequential biosynthesis processes within the ER and Golgi apparatus.

Site-specific chemoenzymatic synthesis is the standard method for producing model glycopeptides, where the initial sugar peptide bond is chemically synthesized. An enzymatic synthesis of the glycan ensues with one sugar residue added at a time[5]. Advances in understanding enzyme specificities and mechanisms have enabled the synthesis of more complex glycopeptides[5], expanding the dimensions of glycopeptide arrays. In vitro methods that produce single, purifiable, and spectroscopically verifiable structures are more suitable than synthetic biology methods since the intention is to measure the kinetics of GTs as well as map out their selectivity and mechanistic action. The localization of GTs, such as GALNTs, has been found to be a regulatory mechanism involved in cancer phenotypes by altering *O*-GalNAc glycan structures

[1]Scientific Computing Research Unit, University of Cape Town, Rondebosch, South Africa. [2]Department of Chemistry, University of Cape Town, Rondebosch, South Africa. [3]Department of Chemistry, Faculty of Applied Sciences, University of Sri Jayewardenepura, Nugegoda, Sri Lanka. e-mail: kevin.naidoo@uct.ac.za

and levels[6,7]. Consequently, in vitro methods must have the capability to mimic the alterations of in vivo glycosylation resulting from the spatial-temporal rearrangement of the distribution and presentation of GTs to the substrate in the ER-Golgi system[8,9].

The first step in mucin-type *O*-linked glycosylation is the addition of GalNAc to serine or threonine residues facilitated by several *N*-acetylgalactosaminyltransferases (GALNTs), forming the Thomsen-nouvelle (Tn) antigen (Fig. 1A). Following this, the addition of galactose to the Tn antigen through T synthase (C1GALT1) to form the T antigen (core 1). Alternative to this, the core 3 can be made by the addition of GlcNAc via β-1,3-*N*-acetylglucosaminyltransferase 6 (B3GNT6).

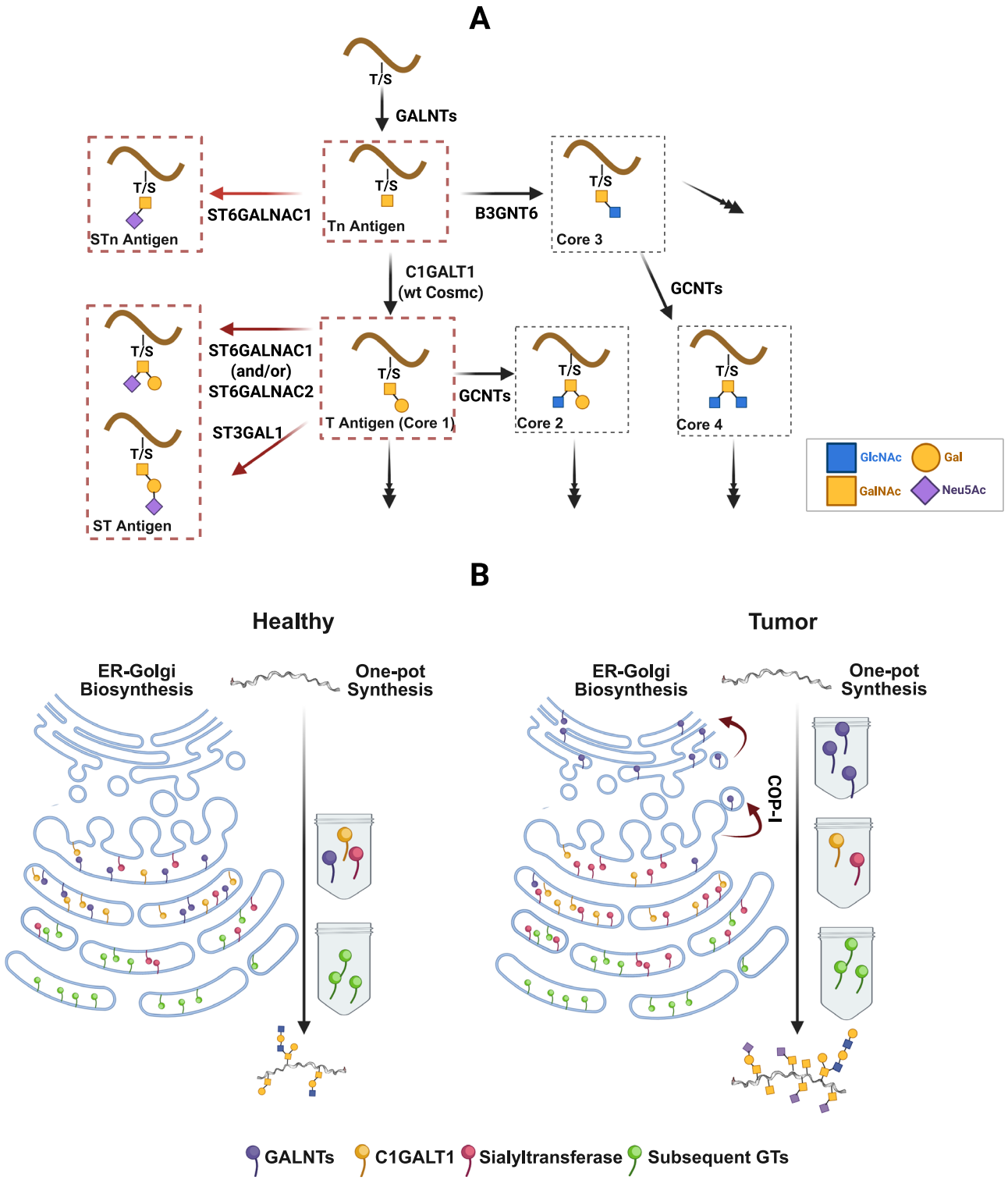


Fig. 1 | Mucin O-GalNAc glycosylation. A O-GalNAc glycosylation pathway showing the reactions that lead to the formation of the cancer-associated Tn and T antigens and their sialylated forms sTn and sT, respectively. **B** Differential localization of GALNTs between normal and tumor epithelial cells is attributed to COP-1-mediated retrograde activation in cancer, shown alongside the corresponding in vitro synthesis design. Deregulation of glycosyltransferase expression is highlighted by tumor-associated upregulation of sialyltransferases. Created in BioRender. Naidoo, K. (2026) https://BioRender.com/kwimrmw.

These foundational structures undergo further branching and elongation with successive additions of monosaccharides such as GlcNAc and galactose, generating diverse glycan chains. Sialyltransferases mediate the sialylation of Tn and T antigens (left pathways in Fig. 1A). These sialylated forms (sTn and sT) terminate chain progression. Clinically, Tn, T, sTn, and sT antigens are significant through their role in establishing the hallmarks of cancer, such as tumor progression, immune evasion, and metastasis[10,11]. The glycosylation of proteins takes place in the ER-Golgi system mostly through glycosyltransferases (GTs) and in some instances in combination with glycosidases. These glycoenzymes are distributed across specific cisternae[8,9,12]. The localization of GTs along the ER-Golgi axis is dynamic, and they are constantly shuffled in both directions via a complex, tightly regulated system involving COP-I and COP-II vesicles[12,13]. This localization across various cisternae has led to an assembly line of compartments performing sequential glycosylation to build glycans on target proteins. In the case of an organism's disease state, a protein's glycan is often altered when there is deregulation of this localization, such as the relocation of GALNTs from the cis Golgi to the ER in tumor formation[6,7]. (Fig. 1B). Although deregulation of GT expression, such as the upregulation of sialyltransferases in tumors, contributes to malignant glycosylation patterns, the underlying mechanisms are unresolved and well beyond the scope of the present work.

Here, we use Mucin 1 (MUC1) as a peptide glycosylation prototype system model. The aim is to build a model that resolves the differences in enzymatic construction between normal glycosylated MUC1 compared with tumor-associated (TA) MUC1. To this end, an in vitro method is developed to simulate ER-Golgi conditions for *O*-GalNAc glycosylation of the MUC1 and TA-MUC1 peptides. Specifically, we illustrate: (i) the kinetic parameters governing glycosyltransferase (GT) activities and substrate specificities using the Universal Glycosyltransferase Continuous assay (UGC assay)[14] that detects nucleoside mono- and di-phosphate formation through nucleotide kinases and NADH consumption, (ii) the molecular mechanisms governing the GTs site specificities, and (iii) the effect of their expression and distribution along the ER-Golgi axis. Following this, the experimental model, along with advanced computer reaction dynamics simulations, was used to reveal peptide site specificity of each of the GTs involved in the synthesis of Tn, T, sTn, and sT antigens at the five unique MUC1 glycosylation sites.

Results

The sequential one-pot synthesis method

To capture the spatial-temporal segregation of glycosylation pathways a fusion tag protein carrying the MUC1 peptide was designed as an assembly conveyor (Fig. 2A). The conveyor vehicle (fusion protein and tags) was tested for biosynthesis interference (Fig. 2B). In the assembly line design for a glycan biosynthesis, the kinetics of the GT-catalyzed reactions and the associated intermediate glycan products are analyzed at every point of construction along the assembly line (Fig. 2C). The data obtained at each point informs subsequent steps, supporting model construction and iterative optimization of the synthesis.

A fusion protein containing the core MUC1 peptide and a carrier protein, superfolder green fluorescent protein (sfGFP), expressed in *E. coli*. sfGFP was selected for its folding efficiency, minimized dimerization, enhanced solubility[15], and tracking the fusion protein during purification, which enables quantification. Central to the design are the work functions of two linkers. A rigid linker (Linker 1) was incorporated to improve steric presentation to the *N*-terminus His-tag, enhancing affinity-based purification (Fig. 1A and Supplementary Fig. 1). To minimize possible interaction with the MUC1 peptide, sfGFP was distanced from the peptide using a second linker (Linker 2) comprising three rigid and three flexible units (from the N to C direction). This provides better accessibility of the *tobacco etch* virus (TEV) protease to its recognition sequence preceding the MUC1 peptide sequence. The design makes possible peptide cleavage at any step of the synthesis while retaining the native MUC1 sequence (Fig. 2A and Supplementary Fig. 1B). A 3D structural model of the fusion protein was generated using AlphaFold2, submitted via the ColabFold Colab notebook (v1.0)[16,17] The model revealed conformational and structural variability around the TEV protease cleavage region confirming the design intention to induce flexibility and as a result accessibility for the protease. In contrast, less conformational variability was predicted around the rigid regions. The carrier protein, along with the linkers and the His-tag, is collectively referred to as the "tag" throughout the manuscript. These features were designed to facilitate in vitro enzymatic glycosylation and enable simple one-step purification (Fig. 2C).

GalNAc-type *O*-glycan (Tn antigen) biosynthesis is initiated by GALNTs, which catalyze α-linkage formation between GalNAc and serine/threonine residues. Each GALNT isoform contains catalytic and lectin domains, enabling three modes of glycosylation: direct, lectin-assisted, or a sequential combination of both. In the lectin-dependent mechanism, the lectin domain anchors to pre-glycosylated sites, guiding the catalytic domain to nearby residues[18–20].

Previously, the specificity of the isoforms GALNT1, 2, and 4 was extensively investigated either using the MUC1 peptide containing multiple tandem repeats or a single repeat with an extension of the N-terminal to the first threonine residue, known as the TAP-24 peptide (Supplementary Fig. 1A)[20–24]. While these constructs revealed GALNT isoform preferences, our aim was to perform kinetic studies that accurately reflect the cumulative contribution of all unique glycosylation sites. We avoid structural redundancy or varying sequence contexts, such as the duplicated T20 site at the N-terminus in TAP-24 or the variable directionality arising from multiple tandem repeats that introduce glycosylation site ambiguity. To achieve this, each site had to be represented only once and sufficiently flanked by the native peptide sequence to preserve catalytic motif recognition. Accordingly, two peptides a 27-mer and a 23-mer were designed by frame-shifting two adjacent repeat units within the MUC1 Variable Number Tandem Repeats (VNTR), as illustrated in Supplementary Figs. 1A and 3A.

The GalNAc acceptor functions of the 27-mer in its fusion form, cleaved form, and the tag alone were tested for each of the GALNT enzymes: GALNT1, GALNT2, GALNT4, and GALNT7 (Fig. 2B). The identical glycosylation rates observed for the tagged and untagged MUC1 confirmed that the fusion tag did not interfere with enzyme activity or substrate specificity. This validation demonstrates that the fusion design preserves the intrinsic glycan-acceptor properties of the MUC1 peptide while providing an optimized carrier for quantitative glycosylation analysis.

In vitro healthy vs. tumor GT distribution and concentration models

The GalNAc-glycosylated sites are subject to either sialylation (addition of Sia) via ST6GALNAC1, galactosylation (addition of Gal) via C1GALT1 and its chaperone C1GALT1C1 (Cosmc), or *N*-acetylglucosaminylation (addition of GlcNAc) via B3GNT6 (Fig. 1A). In healthy contexts, GALNTs and C1GALT1 are localized in the cis-Golgi, while ST6GALNAC1 is distributed across all the cisternae of the Golgi (Fig. 1B)[8,9,12]. No specific localization of B3GNT6 has been reported. The localization of only the GALNTs has been reported to be altered in response to the EGF stimulation of SRC (the proto-oncogene in cancer) via COP-1 mediated retrograde from cis Golgi to the ER[25]. This relocation results in the overexpression of Tn in the ER, where a fraction of this Tn transits to the cell surface without modification, while another fraction transits with modification to T antigen. In patient samples, the same study found that the mean expression of Tn in breast cancer tissue samples was 4.5-fold higher than in normal tissues. From the samples with high Tn expression, 70 % of the samples showed ER localization of GALNTs (inferred indirectly from ER localization of Tn),

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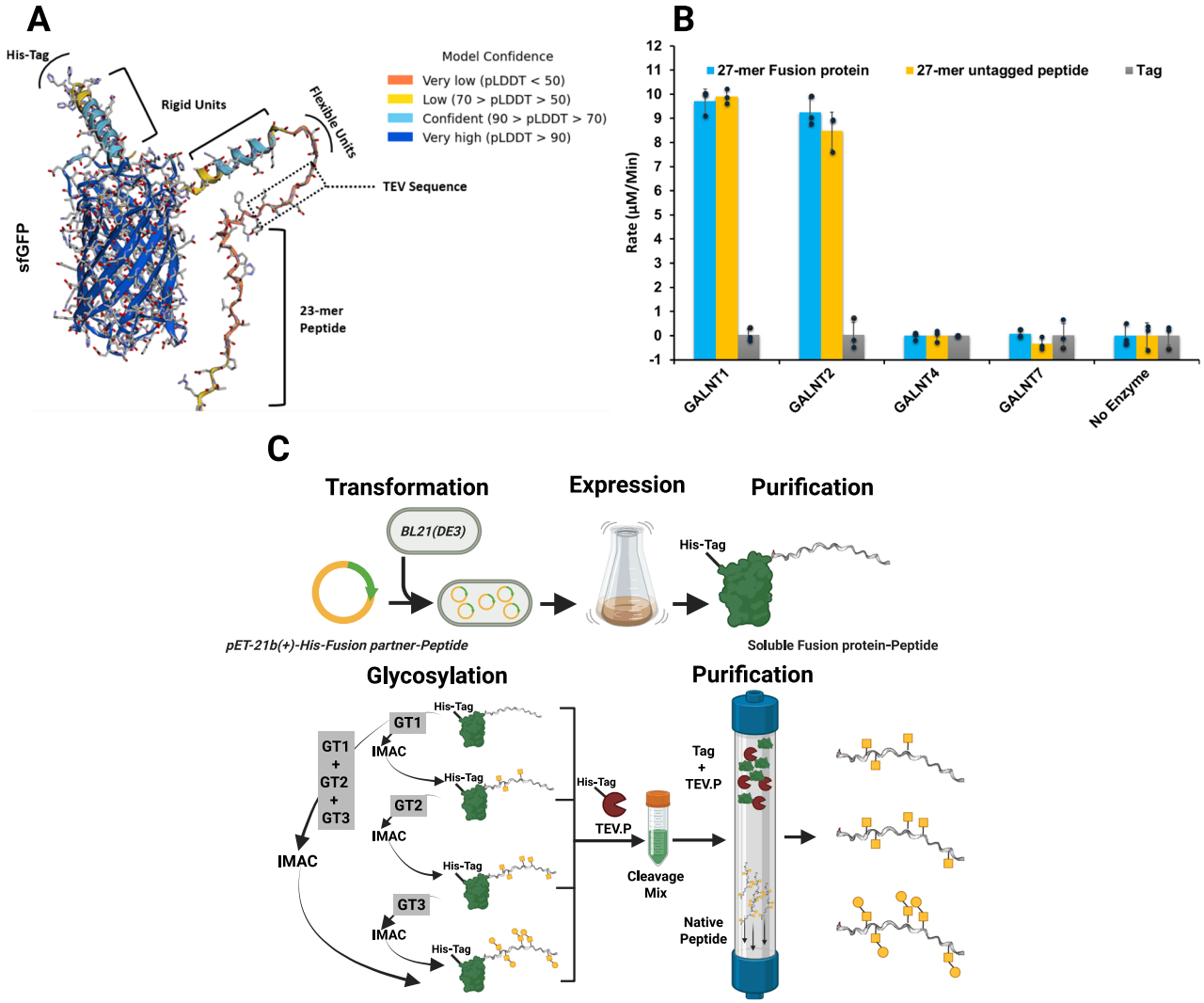


Fig. 2 | The one-pot biosynthesis method. A An AlphaFold structure of the fusion protein for peptide expression. The fusion protein design connected to the MUC1 frame was selected from the peptide tandem repeat. **B** Initial reaction rates for the tag-fused 27-mer MUC1 peptide, the unfused peptide, and the tag-only control as acceptors for GalNAc, catalyzed by various GALNTs. Rates were derived from the slopes of progress curves using the UGC assay. Data are presented as mean ± SD of $n = 3$ technical replicates, with individual data points overlaid. All values are normalized to their respective no-enzyme control reactions. Source data are provided as a Source Data file. **C** The pipeline of expression and enzymatic glycosylation of the MUC1 peptide used to construct the pathway model. It starts with cloning the fusion protein DNA into an expression vector, which is then used to transform *E. coli*. The expressed fusion protein is purified and used in a one-pot glycosylation, performed either stepwise or via multiple glycosylation reactions in a single mixture. The His-tag on the fusion protein makes it possible for a one-step IMAC purification of the glycosylated peptide. The glycosylated peptide is finally cleaved from the fusion protein by TEV protease and collected in the IMAC flowthrough. Created in BioRender. Naidoo, K. (2026) https://BioRender.com/7kjmpbt.

whereas no significant loss of C1GALT1 was detected in these samples, pointing to the ER localization as the driving factor of the observed Tn overexpression[6,7].

Accompanying the relocation of GALNTs, the expression of *O*-GalNAc glycosylation enzymes is altered in cancer compared to normal cells. Furthermore, ST6GALNAC1 was reported to be upregulated in almost all cancer types[26], and C1GALT1 was downregulated, mainly due to Cosmc mutation or epigenetic alteration of both C1GALT1 and Cosmc[27,28]. The objective here is to build a one-pot in vitro synthesis model that will be representative of the impact of the redistribution of GALNTs in altering glycan structures independently of enzyme levels (Figs. 1B and 2C). To achieve this, all activities and kinetics experiments are performed at a standardized enzyme concentration of 100 nM.

The performance of the 23-mer and 27-mer peptides was compared by measuring the relative reactivity of GALNT1, GALNT2, GALNT4, and GALNT7 individually and in various combinations using the UGC assay[14] (Fig. 3A). The primary observation from both experiments is that GALNT4 alone does not react with either peptide, confirming that it relies on a strictly lectin-dependent mechanism. GALNT7 shows no reactivity with either peptide in any combination. On the other hand, GALNT1 and GALNT2 could individually react with both peptides due to their direct catalytic mechanism. The rates of glycosylation via GALNT1 and GALNT2 were generally lower for the 23-mer than the 27-mer. This is explained by the depletion of the direct glycosylation-specific sites T8 and T20 for GALNT1 and GALNT2, respectively, in the 23-mer case, in addition to the absence of secondary lectin-assisted sites in the preferred direction. In contrast, the 27-mer permits the continuation of glycosylation via the lectin-assisted sites (Fig. 3B). When GALNT4 was tested in combination with GALNT1 or GALNT2, using the 27-mer construct, it showed no activity. However, GALNT4 exhibited significant activity using the 23mer peptide when combined with GALNT2, as indicated by the enhanced glycosylation rate when compared with the reaction of GALNT2 individually.

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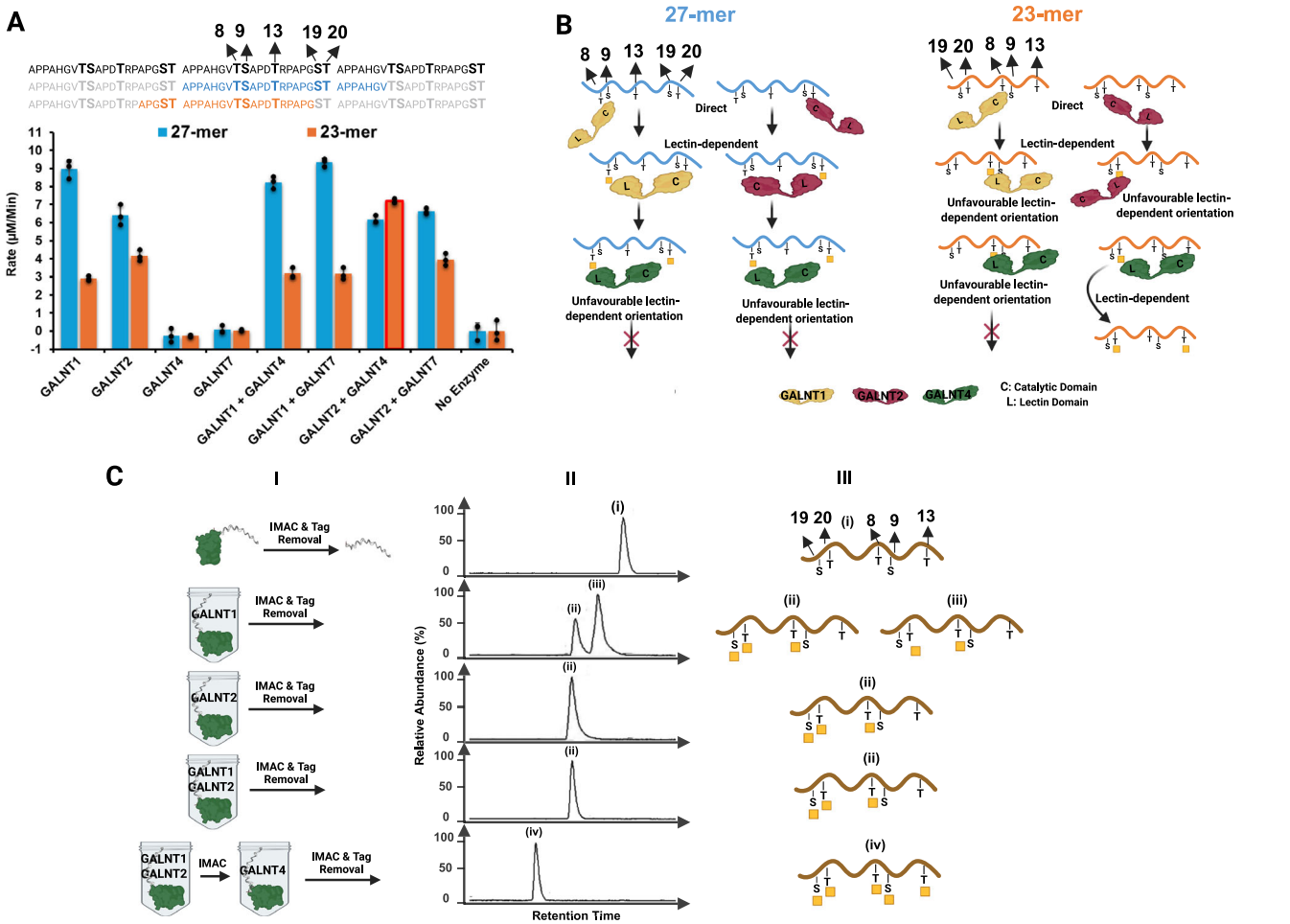


Fig. 3 | *O*-GalNAc glycosylation pathway initiation (synthesis of Tn antigens). **A** Reactivity of the 23-mer and 27-mer peptides with various GALNT enzymes. Initial rates were derived from the linear range of the reaction progress curves. Data are presented as mean ± SD of $n = 3$ technical replicates, with individual data points overlaid. All rates are normalized to their respective no-enzyme control reactions. Attention is drawn to the only instance where GALNT4 in combination with GALNT2 exhibited significantly greater activity on the 23-mer peptide than the 27-mer peptide, through a red emboldened framed bar. Source data are provided as a Source Data file. **B** A schematic illustration of the direct and lectin-dependent mechanisms of the glycosylation reactions of GALNT1, GALNT2 and GALNT4 with the 23-mer and the 27-mer peptides. **C** In vitro synthesis of GalNAc-glycosylated 23mer: I. Brief description of synthesis designs and steps. II. Abundance of the products of each reaction as measured by liquid chromatography. Source data are provided as a Source Data file. III. Structures of the products of each reaction, as determined from the LC-MS analysis and the synthesis pathway, lead to these structures, as concluded from the preceding Data. Details of the LC-MS data and analysis are given in the Supportive Information (Supplementary Figs. 4–9). Created in BioRender. Naidoo, K. (2026) https://BioRender.com/d5imyhz.

GALNT2 is known to have a preferred specificity to T20 (in the PGST sequence) via direct catalytic domain recognition. In the 23-mer, T20 is at the *N*-terminal to T13 and S9 (GALNT4 specific sites), and pre-glycosylation at T20 left to T13 and S9 sites is the prerequisite for GALNT4 lectin-dependent specificity (Fig. 3B)[21]. The collective evidence presented here confirms the efficiency of the fusion protein design to mimic natural MUC1, while its tag components do not affect the catalytic function of the GALNTs. Additionally, 23-mer has proved valid to capture the reaction mechanism of GALNT4 and the direct mechanism for both GALNT1 and GALNT2. However, the reported lectin-dependent mechanisms for GALNT1 and GALNT2 can be better studied using the 27mer.

In vitro synthesis of the Tn antigen

In vitro synthesis of the Tn antigen was carried out to explore the possible structures enabled by extensive glycosylation (higher enzyme concentrations for a longer period) using combinations of GALNTs that demonstrated activity in the UGC assay (Fig. 3A). Since it was established that GALNT4, as well as GALNT7, did not show activity to initiate glycosylation on the naked peptide, they were excluded from the synthesis experiments (Fig. 3C(I)). LC-MS results confirmed the structure and purity (Supplementary Fig. 5) of the peptide (i) observed from a single liquid chromatography peak (Fig. 3C (II)). The catalytic action of GALNT1 results in glycosylation of 2 or 3 sites, and 3 sites in the case of GALNT2 (Fig. 3C; Supplementary Figs. 6 and 7). The glycosylation of T8 and T20 is consistent with a preferred direct GALNT1 and GALNT2 mechanism, respectively. However, the additional glycosylation sites observed beyond the primary GALNT1 (T8) and GALNT2 (T20) preferences do not align with the previously reported lectin-dependent mechanisms for these enzymes on MUC1. Instead, glycosylation of these secondary sites likely reflects a less specific mode of glycosylation[21,23]. This is evident from short-term recorded activity on the remaining sites (Fig. 3A) because the 23mer is not structurally disposed to the directionality required for this mechanism, whereas the 27-mer has the needed directional sequence. A randomized peptide sequence study found that GALNT1 can catalyze a long-range N-terminal lectin-dependent mechanism, while GALNT 2 can participate in both N and C long-range lectin-dependent mechanism[29]. Another study confirmed the bi-directionality of the lectin-dependent mechanism for both enzymes, with a preference

given to N and C long-range lectin-dependent mechanisms for GALNT1 and GALNT2, respectively[20]. Whether the additional two sites represent less-preferable direct glycosylation or lectin-dependent events remains to be proven. While this can potentially be resolved by mutating the lectin domains, such an investigation is outside the focus of the biosynthesis one-pot model being developed here. No further glycosylation beyond three sites was observed when GALNT1 and GALNT2 were combined (Fig. 3C and Supplementary Fig. 8), suggesting an absence of specificity from the two enzymes toward the two remaining sites S9 and T13.

When the product resulting from GALNT1 and GALNT2 activity was glycosylated with GALNT4, all five sites were glycosylated (Fig. 3C and Supplementary Fig. 9). This indicates that the remaining two sites, S9 and T13, are GALNT4-specific and are glycosylated through a lectin-dependent mechanism. The consensus view from all previously reported results is that T8 and T20 are specific sites for GALNT1 and GALNT2, respectively, while S9 and T13 are GALNT4 specific. Taken together, it can be concluded that GALNT1, GALNT2, or their combination can glycosylate S19, T20, and T8 by utilizing direct and lectin-dependent mechanisms. However, GALNT1 is less efficient than GALNT2 in completing the glycosylation of either S19 or T20 (Fig. 3C (III)).

It is widely accepted that the GalNAc site occupancy increases with GALNTs overexpression and their ER relocation[6,7]. The effect of these two factors was simulated here in the one-pot biosynthesis model, where extensive glycosylation was performed to interrogate the action of GALNTs in isolation from other GTs. It was seen that sites, such as T20, S19, and T8, were not selective and can be glycosylated by multiple GALNTs, such as GALNT1 and GALNT2, as shown here, as well as GALNT3, which was previously reported[21,23]. We now refer to these sites that are the first ones to be glycosylated as GALNT_D-sites. The S9 and T13 sites are strictly lectin-dependent and thus are dependent on prior glycosylation of T20, S19, and T8, which we refer to now as GALNT_L-sites (Fig. 4A). The lectin-dependent mechanism is therefore responsible for high-density GalNAc-*O*-glycosylation[24]. This differential site occupancy was also found to be associated with cancer transformation. The site saturation was observed in MUC1 expressed in tumor cells compared with normal MUC1 in breast milk[30]. The GalNAc3-23mer (3 glycosylated sites at GALNT_D-sites only) and GalNAc5-23mer (5 glycosylated sites at GALNT_D-sites and GALNT_L-sites), synthesized here, will be used as models for site occupancy, semi-glycosylated and completely-glycosylated, respectively.

In vitro synthesis of the T and sTn antigen and core 3

The sequential addition of one or two sugars to the Tn antigen generates diverse glycan core structures that yield eight distinctly different cores. Cores 1–4 are the most common structures (Fig. 1A) while cores 5–8 are rare[31]. Core 1, also known as the T antigen, is formed by the addition of a galactose residue in a β1-3 linkage to the Tn antigen. This is achieved by T synthase (C1GALT1) in association with its chaperone C1GALT1C1 (Cosmc). Core 3 is synthesized by B3GNT6, which catalyzes the GlcNAc β1-3 to Tn antigen bond. The ST6GALNAC1 sialylates the Tn antigen, producing sialyl-Tn (sTn) that terminates further glycosylation. Of specific interest here, C1GALT1 or B3GNT3 catalysis is prevented. Core 2 and core 4 are produced by extending core 1 and core 3, respectively, through the action of GCNT enzymes, and can be further glycosylated, producing more complex structures (Fig. 1A).

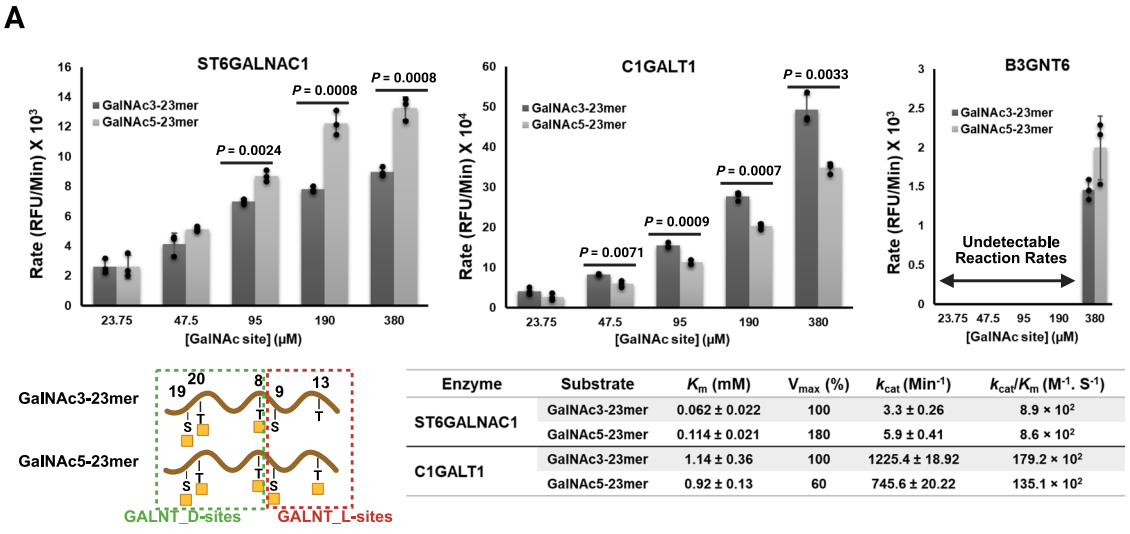
The reactivity of ST6GALNAC1, C1GALT1, and B3GNT6 was compared for GalNAc3-23mer (semi-glycosylated) and GalNAc5-23mer (completely-glycosylated) as models for site occupancy. Serial dilutions of both substrates were prepared by normalizing the concentrations to the number of GalNAc-glycosylated sites (Fig. 4A). While the overall activity of C1GALT1 was much higher than that of ST6GALNAC1, as indicated from k_{cat} values (table in Fig. 4A), the results show that ST6GALNAC1 preferably sialylates the GalNAc5-23mer MUC1, whereas C1GALT1 preferably glycosylates the GalNAc3-23mer MUC1. The difference in the selectivity of both C1GALT1 and ST6GALNAC1 increases with an increase in MUC1 concentration. The kinetics parameters derived from the dose-response of ST6GALNAC1 for both semi- and completely-glycosylated MUC1 indicates that the enzyme has a lower affinity (K_m value of 0.114 mM for GalNAc5-23mer vs. 0.062 mM for GalNAc3-23mer) but higher turnover (Vmax for GalNAc5-23mer is 180 % of the Vmax of GalNAc3-23mer) of the fully glycosylated MUC1 compared with the semi glycosylated MUC1 (table in Fig. 4A). On the other hand, no significant difference in the K_m values of C1GALT1 were observed between the semi-saturated and completely saturated MUC1. While these kinetic parameters serve as proxies for relative enzyme activity on defined substrates, the insights they provide are inherently limited. In particular, GT performances under physiological conditions are influenced by multiple factors, including enzyme membrane association, response to in vivo complex substrate structures, and intracellular enzyme and substrate concentrations and fluxes. All of these can be studied in isolation or in combination to arrive at a comprehensive narrative on GT site preference and activity. Notwithstanding the activity of ST6GALNAC2 (validated with asialofetuin, a model acceptor for 2,6-sialylation[32]), no significant glycosylation was detected on either substrate.

Evaluating ST6GALNAC1 Tn site specificity

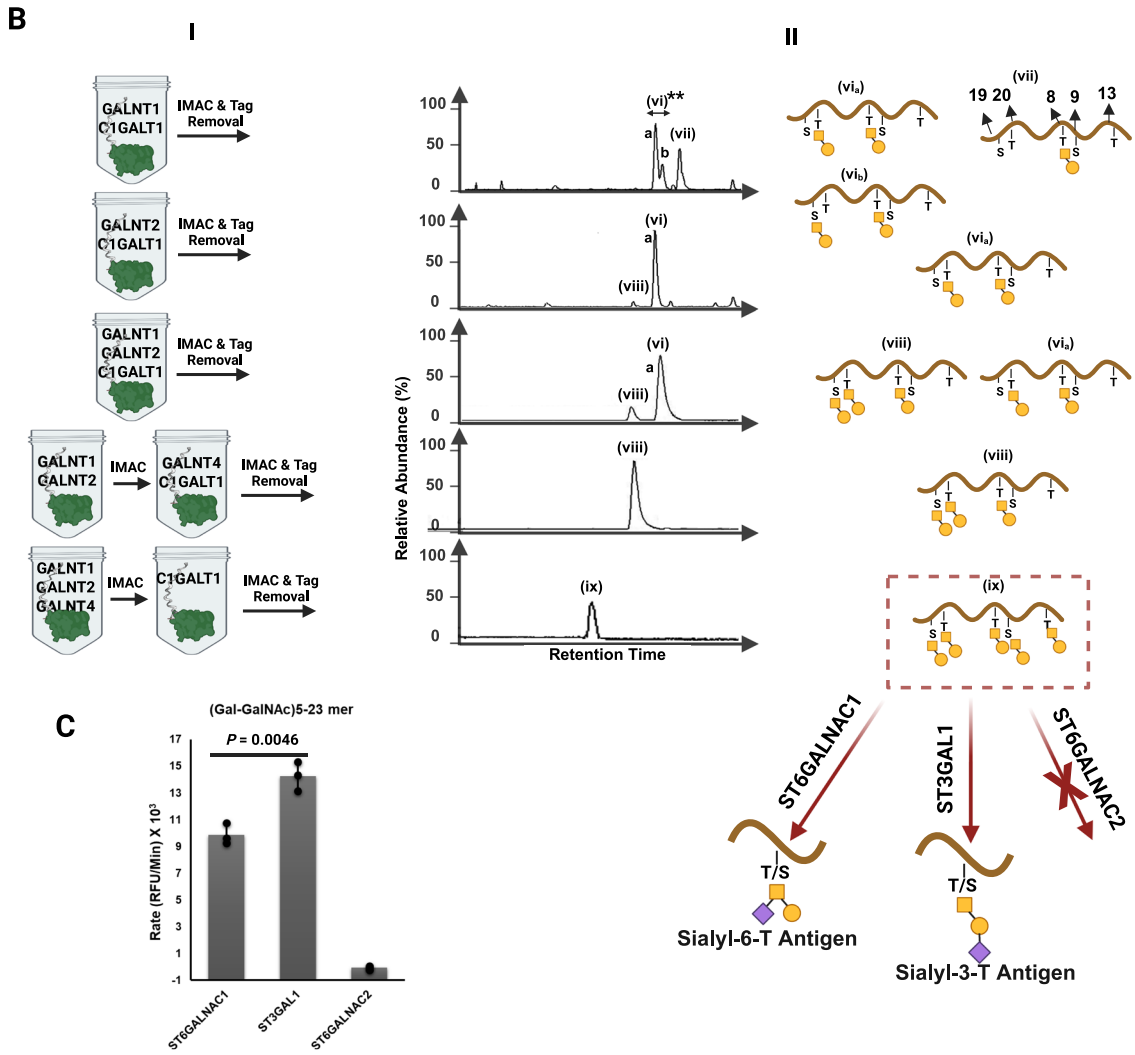
The formation of the sTn MUC1 is a key antigen in several cancers; consequently, a detailed molecular description of the location of this epitope on the MUC1 frame is essential for drug discovery as well as vaccine development. In the section detailing Michaelis-Menten kinetics experiments (Fig. 4A), it was revealed that ST6GALNAC1 prefers glycosylation of the GALNT_L-sites over the GALNT_D-sites of MUC1. A computational study was performed to investigate the sialylation of the five possible glycosylation sites of the cancer-associated, fully glycosylated (GalNAc5-23mer) MUC1 peptide.

The initial peptide to enzyme binding positions were determined from protein-protein docking of double tandem repeat (DTR)-MUC1 to the ST6GALNAC1 enzyme. The DTR-MUC1 was used to mitigate possible binding biases that may arise from the conformationally flexible terminal ends of the peptide. Following this, the single tandem repeat (STR)-MUC1 23-mer (23-mer) peptide sequence was docked, and no differences in comparison to the DTR-MUC1 for binding preferences were observed (Supplementary Fig. 23 I, II, and III). The lowest energy pose from STR-MUC1-ST6GALNAC1 docking reveals an orientation of the region containing T20 and S19 closest to the catalytic pocket. A comparison of the center of mass (COM) MUC1 glycosylation site to the COM of the CMP-Neu5Ac donor for T20 (17.54 Å) and S19 (18.56 Å) shows a favourability for the former. However, the largest cluster of produced poses placed the PDTR region, specifically T13, closest to the catalytic pocket. Here, the T13 is at a distance of 13.66 Å from its COM to the COM of the CMP-Neu5Ac donor and a distance of 9.72 Å from the COM of T13 to the COM of the catalytic base His567 (Supplementary Table 2). None of the poses generated placed the region containing S9 and T8 directly over the catalytic domain. Previous studies have shown that it is possible for ST6GALNAC1 to sialylate T8 in addition to T20 and T13. Further it has been shown that ST6GALNAC1 has negligible activity on MUC1 serine residues[33]. To account for the known T8 reactivity, an additional STR-MUC1-ST6GALNAC1 docking experiment was undertaken to produce a complex that places the APGSTAPPAHGV region containing S9 and T8 in closer proximity to the catalytic domain. This generated a pose which showed that T8 COM is closer (19.95 Å) than the S9 COM (23.88 Å) to the COM of the CMP-Neu5AC donor. In both binding poses featuring adjacent T and S sites, i.e., T20 and S19 and T8 and S9, the threonine residues are orientated closer to the CMP-Neu5Ac donor. This supports previous

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Enzyme	Substrate	K_m (mM)	V_{max} (%)	k_{cat} (Min^{-1})	k_{cat}/K_m ($M^{-1}.S^{-1}$)
ST6GALNAC1	GalNAc3-23mer	0.062 ± 0.022	100	3.3 ± 0.26	8.9×10^2
	GalNAc5-23mer	0.114 ± 0.021	180	5.9 ± 0.41	8.6×10^2
C1GALT1	GalNAc3-23mer	1.14 ± 0.36	100	1225.4 ± 18.92	179.2×10^2
	GalNAc5-23mer	0.92 ± 0.13	60	745.6 ± 20.22	135.1×10^2



observations that ST6GALNAC1 preferably sialylates threonine over serine.

QM/MM reaction dynamics simulations were performed to provide a qualitative mechanistic understanding of the experimentally observed trends. The reaction dynamics simulations were modeled using the CHARMM force field to calculate the classical Hamiltonian and SCCDFTB to calculate the quantum Hamiltonian. Simulations were performed with identical simulation conditions to provide qualitative, comparative mechanistic insight into site-specific sialylation rather than to derive absolute kinetic or energetic parameters. The QM region comprised His567 (catalytic base) sidechain atoms, His552 (stabilizing the CMP phosphate group), the Neu5Ac moiety, the CMP phosphate group and the GalNAc acceptor. The boundaries were treated with link atoms using Hydrogen link atoms for nonpolar C-C bonds and the Simple Link Atom Saccharide Hybrid (SLASH)[34] method for polar C-O bonds. Reaction dynamics simulations were carried out to compare

Fig. 4 | Assessment of the first layer of the structure extension after Tn synthesis. A Effect of GalNAc site occupancy on the specificity of ST6GALNAC1, C1GALT1 and B3GNT6. Serial dilution of concentrations of GalNAc3-23mer and GalNAc5-23mer was calculated per GalNAc site occupancy. Data is presented as mean ± SD, of $n = 3$ technical replicates, with individual data points overlaid. Kinetics parameters are reported as apparent values, due to the presence of multiple reactive sites on the substrates; therefore, they reflect the combined contribution of all the sites[52]. V_{max} values are presented as a percentage relative to GalNAc3-23mer for each enzyme. Kinetics values in the table represent the best-fit parameter ± SE. Source data are provided as a Source Data file. **B** Core 1 Synthesis: Effect of GALNTs vs. C1GALT1 competition on site occupancy. I. Synthesis designs of different combinations of sequential synthesis reflecting C1GALT1 competition with GALNTs in isolation or in different combinations. II. shows the LC peaks and structures deduced from analyzing mass spectrometry results (Supplementary Figs. 10–14, and source data are provided as a Source Data file).** Peak vi corresponds to the incorporation of two GalNAc molecules, as indicated from the LC-MS analysis. **C** Sialylation of core 1 via ST3GAL1, ST6GALNAC1, and ST6GALNAC2. Data is presented as mean ± SD, of $n = 3$ technical replicates, with individual data points overlaid. Source data are provided as a Source Data file. For panels (**A**, **C**), statistical significance was determined using a two-tailed, two-sample equal variance Student's *t*-test. Only *P*-values < 0.01 are displayed on the graph; comparisons not labeled were not statistically significant. Created in BioRender. Naidoo, K. (2026) https://BioRender.com/ps0u8gp.

the free energy reaction profiles along the minimum energy pathways (MEPs) for ST6GALNAC1 catalyzed sialylation on the T20, T8 and T13 sites and to investigate the site-specific differences between them. Free energy simulations were conducted along two reaction coordinates to monitor bond formation and breaking using the Free Energies of Adaptive Reaction Coordinate Forces (FEARCF)[35–37] method. Minimum energy pathways (MEPs) were determined from the resulting free energy surfaces calculated by FEARCF. These MEPs are demonstrated in the form of 1D plots indicating the change in free energy of the reaction across a linear combination of bond forming (C2-O6) and bond breaking (C2-O2) primary reaction coordinates.

QM/MM reaction dynamics simulations showed the ST6GALNAC1 sialylation reaction is characterized by the cleavage of the CMP-Neu5Ac C2-O2 bond and the formation of a stable Oxocarbenium intermediate (OC). This is facilitated by hydrogen bond formation between the phosphate moiety and His552. Product formation occurs when the GalNAc O6 hydroxyl is deprotonated by the catalytic base, His567, enabling nucleophilic attack on the Neu5Ac C2. While this reaction mechanism (Fig. 5A) is common to all three sites, QM/MM simulations revealed essential differences between them.

To quantify site-specific differences, three average distances were computed from the reaction trajectories: the distance between the GalNAc C6 and the CMP-Neu5Ac C2 anomeric carbon (r1); the distance between GalNAc C6 and NE2 of the catalytic base His567 (r2), and the distance between O2P of the CMP moiety and the His552 HE2 (r3). The angle formed between the His552 HE2, O2P and the CMP phosphorus atom was also measured and tabulated (Fig. 5B). These metrics were calculated using GalNac C6 instead of O6 since the hydroxyl group shows high rotational freedom, leading to large positional changes. C6 anchors the O6 hydroxy involved in the reaction and provides a more stable reference point for evaluating the acceptor position relative to the other catalytically relevant residues. The T20 system showed the smallest C6-C2 distance (4.28 ± 0.59 Å) followed by T13 (4.59 ± 0.61 Å) and T8 (4.86 ± 0.57 Å). A similar trend was seen in the C6-NE2 distances with T20 exhibiting the closest contact with an average distance of 3.71 ± 0.59 Å, followed by T13 (4.57 ± 0.71 Å) and T8 (4.17 ± 0.95 Å). However, on closer inspection of the geometry needed for a successful catalysis, the T13 system had the shortest average O2P-HE2(His552) distance of 1.73 ± 0.23 Å necessary to surmount the TS2 barrier. In the QM/MM runs, T20 and T8 displayed greater average distances (1.99 ± 0.54 Å and 2.11 ± 0.79 Å respectively). A comparison of the P-O2P-HE2(His552) angle for the three cases showed that T13 displayed a more linear angle of 133.65 ± 18.41°, enabling electron diffusion along the bonds compared with the less linear angles seen in T20 and T8 (113.12 ± 23.79° and 114.10 ± 20.89°). In contrast, T8 showed the largest average O2P-HE2(His552) distance as well as the largest standard deviation in both the O2P-HE2(His552) distance and the P-O2P-HE2(His552) angle. This instability of the T8 system is expected to impede the formation of the OC intermediate, increasing the barrier of TS1. In summary of the three systems, T13 showed the smallest standard deviation for both the O2P-HE2(His552) distance and the P-O2P-HE2(His552) angle, indicating an enduring approach to the His552 catalytic residue. This more favorable orientation in the T13 case implies that the cleavage of the C2-O2 bond, leading to the formation of the oxocarbenium (OC) intermediate with a favorable TS1 energy barrier.

T20 and T13 share similar distances between the GalNAc acceptor and Neu5Ac donor (C6-C2) as well as between GalNAc and the catalytic histidine (C6-NE2). We therefore expect similar TS2 values associated with the conversion of the oxocarbenium (OC) intermediate to the products for these sites. In comparison, T8 displayed larger distances across these metrics, making it a less favorable setup for product formation and, therefore, a higher TS2 energy barrier compared to T20 and T13 (Fig. 5C).

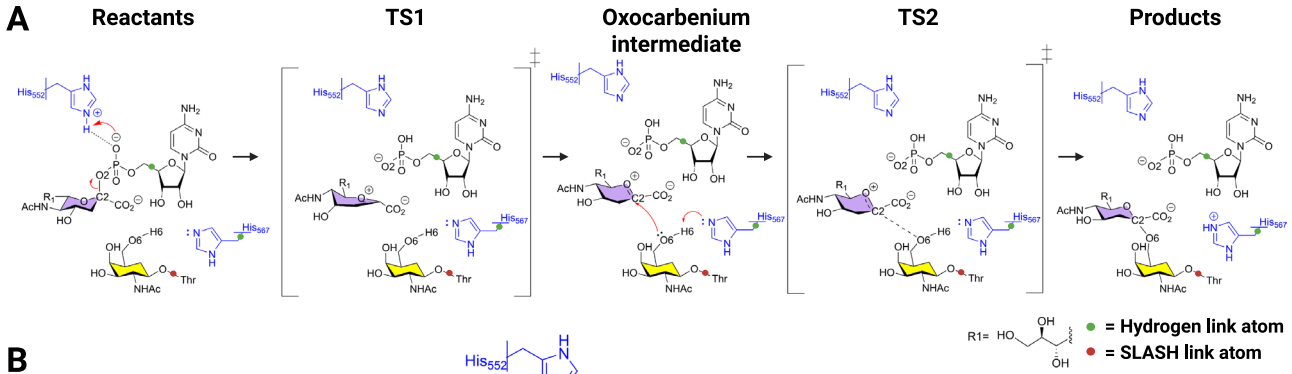
The calculated MEPs are consistent with the desiccation-driven mechanism[38]. In the case where Tn is sialylated at the T8, product formation was observed following transition state energy barriers of 20.66 kcal/mol (TS1) and 23.87 kcal/mol (TS2). The sialylation of T13 proceeded via a TS1 energy barrier of 15.12 kcal/mol to form a stable oxocarbenium intermediate (OC) with an energy of 11.91 kcal/mol. The formation of products was observed after overcoming the TS2 energy barrier of 16.73 kcal/mol. Of the three simulated reactions, T8 shows the highest transition state barrier (23.87 kcal/mol) and therefore the slowest rate, making this site the least favored. The sialylation of the T20 and T13 residues exhibited similar energetic profiles with only a 0.38 kcal/mol difference in TS2 values. This suggests that sialylation at the T13 site may be slightly more favored over sialylation at the T20 site.

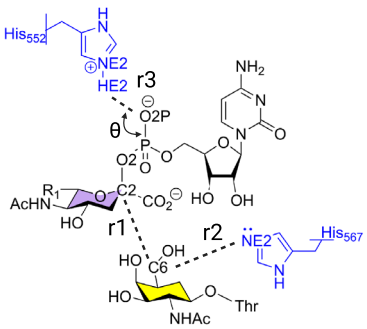
A combined assessment of the distance analysis from the docked DTR-MUC1 and STR-MUC1 poses, the QM/MM trajectory analysis of the catalytic binding dynamics and the MEP pathways derived from QM/MM FEARCF reaction dynamics simulations, leads to the conclusion that initial sialylation of GalNAc5-MUC1 favors the T13 site in the GALNT1_L region (not S9) over the GALNT1_D sites S19, T20, and T8. While QM/MM simulations provide valuable mechanistic insight, their role in this study is supportive, and conclusions are grounded primarily in kinetic and LC–MS experimental data.

In vitro synthesis of core 1 (T antigen)

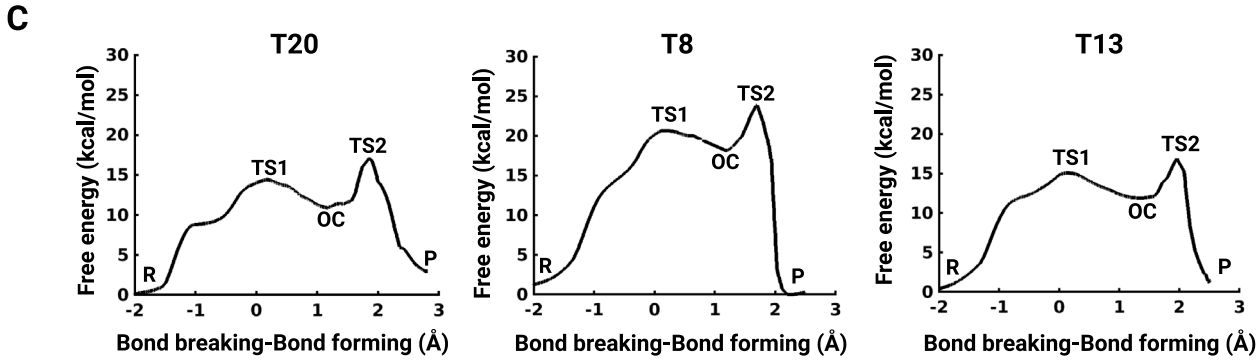
The synthesis of core 1 via C1GALT1 was performed using different one-pot designs to simulate the varying distribution of GALNTs across the ER-Golgi system between healthy and tumor settings. Two scenarios were modeled: (1) when C1GALT1 competes with GALNTs, mimicking their co-localization in the cis-Golgi, and (2) when GALNTs act on the peptide first in isolation, representing their re-localization to the ER. When C1GALT1 was combined with GALNT1 (Fig. 4B(I)), the chromatogram displayed three peaks (vi_a, vi_b, and vii, Fig. 4B(II) and Supplementary Fig. 10), indicating the incorporation of Gal-GalNAc at one or two sites. Two of these peaks via and vi_b are partially overlapping and share the same m/z value for their precursor ions, consistent with the incorporation of two Gal-GalNAc residues. Since T8 is the primary site for GALNT1, and S9 and T13 are specific to GALNT4, the single-site product is glycosylation at T8. The isomers vi_a and vi_b can therefore be explained by a second glycosylation event occurring at either S19 or T20.

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Metric	T20	T8	T13
r1[C6-C2] (Å)	4.28 ± 0.59	4.86 ± 0.57	4.59 ± 0.61
r2[C6-NE2(His567)] (Å)	3.71 ± 0.59	4.57 ± 0.71	4.17 ± 0.95
r3[O2P-HE2(His552)] (Å)	1.99 ± 0.54	2.11 ± 0.79	1.73 ± 0.23
θ[P-O2P-HE2(His552)] (°)	113.12 ± 23.79	114.10 ± 20.89	133.65 ± 18.41



Metric	T20	T8	T13
TS1 (kcal/mol)	14.38	20.66	15.12
TS2 (kcal/mol)	17.11	23.87	16.73

Fig. 5 | Computational modeling of the reaction site specificity of ST6GALNAC1. **A** Proposed S_N1-like reaction mechanism of ST6GALNAC1 elucidated from QM/MM reaction dynamic simulation. The position of link atoms added to prepare the system for QM/MM simulations is marked in green for Hydrogen link atoms and in red for Simple Link Atom Saccharide Hybrid (SLASH) link atoms (see section "Computational methods"). Enzyme amino acids are drawn in blue with GalNAc and Neu5Ac rings filled in yellow and purple, respectively. **B** 2D schematic indicating key distances and angles that influence reactivity. Average C6-C2, C6-NE2, O2P-HE2 distances, and NE2–C6–C2 angle values across the QM/MM simulation for each system can be found tabulated below. **C** 1D representation of the minimum energy pathway determined from the free energy surface showing the free energy associated with the bond forming and bond breaking in going from the reactants (R) to the products (P) via two transition states (TS1 & TS2) and an oxocarbenium intermediate (OC). Transition state barrier heights are summarized in a table below the 1D plots. 3D Molecular representations of key chemical species along the reaction pathway (R, TS1, OC, TS and P) are provided in Supplementary Fig. 24. Source data are provided as a Source Data file. Created in BioRender. Naidoo, K. (2026) https://BioRender.com/v8q23hm.

When C1GALT1 was combined with GALNT2, the reaction predominantly yielded a uniform two-site product with the same LC retention time as the via peak from the GALNT1 + C1GALT1 reaction (Fig. 4B and Supplementary Fig. 10). A recent study on MUC1 substrate specificities showed that T20 and T8 are efficiently glycosylated by both GALNT1 and GALNT2, whereas S19 is glycosylated much less efficiently[39]. Based on this, the more predominant product via corresponds to glycosylation at T8 and T20, whereas vi_b corresponds to glycosylation at S19 and T20. In the one-pot biosynthesis mix containing GALNT1, GALNT2, and C1GALT1, a major peak corresponding to a peptide with two Gal-GalNAc modifications was observed (Fig. 4B and Supplementary Fig. 11). These were assigned to T8 and T20, the

primary sites for GALNT1 and GALNT2, respectively, further supporting the assignment of vi. Along with this, a small proportion of products(viii) bearing Gal-GalNAc at all three sites (S19, T20, and T8) were detected in the one-pot synthetic mix that included GALNT2. The amount was significantly smaller compared to a synthetic mix that excluded C1GALT1 (Fig. 3C).

These findings demonstrate that GALNT1 and GALNT2 mixed with C1GALT1, i.e., (GALNT1, C1GALT1), (GALNT2, C1GALT1) or (GALNT1, GALNT2, C1GALT1) reduces site occupancy by one site compared to when they act alone. This suggests that after GalNAc is added to T8 or T20 by the direct mechanism of GALNT1 or GALNT2, respectively, C1GALT1 competes with these enzymes' lectin domains for the modified sites. The lectin domains of GALNTs are specific for monosaccharide GalNAc binding, as displayed in the Tn antigen and have no binding affinity to sugars displayed on disaccharides or oligosaccharide glycans (such as in sTn, sT, or T)[40]. Consequently, once C1GALT1 catalyzes the addition of Gal onto GalNAc to yield a core 1 structure, further GalNAc additions facilitated by the lectin-dependent mechanism of GALNTs are inhibited. An additional factor may contribute to the reduced site occupancy observed with the reaction of GALNT1 in combination with C1GALT1. In this reaction, two isomers were detected that differ in glycosylation of adjacent sites, being either S19 or T20. This suggests that after initial GalNAc addition by GALNT1, extension of the structure to GalNAc-Gal by C1GALT1 could create steric hindrance, preventing GALNT1 from glycosylating the neighboring site. This effect appears less prominent for GALNT2, as reactions containing GALNT2 still show occupancy at all three sites.

In line with this, revealing and quantifying the exact contributions of lectin-domain inhibition vs. the proposed steric hindrance would require a more rigorous investigation, potentially involving mutagenesis of the GALNT lectin domains and site-specific LC-MS/MS analysis, such as Electron Transfer Dissociation (ETD) for precise O-glycan site localization. Analysis such as this is out of the scope as it distracts from the focus of the present work to model the system of glycosylation biosynthesis within normal or diseased state Golgi.

Finally, when simulating the ER localization of GALNTs (i.e., when the combined GALNT1, 2 and 4 product is followed by incubation with C1GALT1), C1GALT1 generated five sites occupied by core 1 structures (Fig. 4B and Supplementary Fig. 13). The impact of GALNT co-localization with C1GALT1 on site occupancy can be extended to their co-localization with ST6GALNAC1. This is supported by a study showing that overexpression of ST6GALNAC1 reduced site occupancy by 25% in Chinese Hamster Ovary (CHO) cells[12].

Core 1 Sialylation

Core 1 can be sialylated via ST3GAL1 to form the sialyl-3-T antigen or via the ST6GALNAC family to form the sialyl-6-T antigen (Fig. 4C). However, despite the reported activities of the three enzymes on the T antigen, the specificities of these enzymes were not tested comprehensively using standardized substrates. Additionally, details of ST6GALNAC1 vs. ST6GALNAC2 specificities for Tn and T are conflicting across studies and the models (in vitro vs. in vivo)[41,42]. Both ST6GALNAC1 and ST6GALNAC2 show no reactivity with the stand-alone GalNAc or Gal-GalNAc acceptor, which confirms the peptide core requirement for both enzymatic activities[41,42]. That was confirmed when both showed significant reactivity with asialofetuin (Supplementary Fig. 25). We showed above that ST6GALNAC1 (but not ST6GALNAC2) reacts with Tn antigen. The same observation was true for the T antigen (Fig. 4C). When compared to ST6GALNAC1, ST3GAL1 showed significantly higher reactivity with the T antigen; the results suggest that sialyl-3-T is more predominant than sialyl-6-T.

DTR-MUC1 One-pot biosynthesis model of GALNT localization

To verify that the observations made from STR-MUC1 23-mer peptide in vitro synthesis were not context dependant, the model one-pot synthesis was redone on a DTR-MUC1 sequence (Fig. 6A). Each glycosylation site was uniquely represented in the STR-MUC1 23-mer structure making unambiguous kinetic observations possible and in so revealing a clear mechanistic distinction between direct and lectin-dependent initiation by GALNT1, GALNT2, and GALNT4. However, extrapolation to a multimer VNTR-MUC1 from this STR-MUC1 model is not possible as it fails to account for the influence of the directionality of the effect of adjacent repeats on lectin-mediated glycosylation in multiple tandem repeats found in MUC1. To achieve a more physiologically relevant simulation, the double-repeat MUC1 peptide accounted for the STR-MUC1 shortcomings and was used to model two conditions: (i) the Golgi colocalization of GALNTs with downstream glycosyltransferases (C1GALT1 and ST6GALNAC1), representing normal tissue glycosylation (Fig. 6), and (ii) the tumor-associated relocalization of GALNTs to the endoplasmic reticulum (Fig. 7).

Colocalization of GALNTs with C1GALT1/ST6GALNAC1 in Normal Golgi

The DTR-MUC1 peptide was expressed and purified as a fusion protein following the same procedure described for the 23-mer peptide (see Methods). The peptide was cleaved using TEV protease, and the resulting product was confirmed by LC–MS analysis (Fig. 6A and Supplementary Fig. 14). To model Golgi colocalization of GALNTs with downstream glycosyltransferases, two one-pot synthesis experiments were carried out. In each, GALNT1, GALNT2, and GALNT4 were incubated with either C1GALT1 or ST6GALNAC1 to independently examine the influence of each enzyme on site occupancy. The products of each experiment were purified and subjected to digestion with IMPa, a metalloprotease that specifically cleaves N-terminal to O-GalNAc glycosylation sites, except when the modified residue is preceded by an aspartic acid. This selective cleavage pattern facilitates the mapping and diagnosis of site occupancy across the glycopeptide[43]. Both reaction experiments yielded a predominance of glycosylation at T8 and T20, with only minor or trace occupancy at S19. The GALNT_L sites (S9 and T13) were not glycosylated. These findings reproduce the site-specific pattern observed with the STR-MUC1 peptide, confirming that for conditions mimicking normal Golgi colocalization, glycosylation is primarily directed to GALNT_D sites, with minimal contribution from lectin-dependent mechanisms.

Relocation of GALNTs in Tumor-associated ER

The modeling of GALNT relocalization of to the ER (Fig. 7A), was done by evaluating their activity as (GALNT1, GALNT2) or in combination with GALNT4 (GALNT1, GALNT2, GALNT4). Exposure of the DTR-MUC1 peptide to GALNT1 and GALNT2 resulted in complete glycosylation of the GALNT_D sites, S19, T20, and T8, while the GALNT_L sites were not glycosylated. This confirms the limited specificity of GALNT1 and GALNT2 for GALNT_D sites and shows that the presence of an additional tandem repeat in the form of DTR-MUC1 at either terminus does not enable lectin-dependent glycosylation at GALNT_L sites. The inclusion of GALNT4 to the GALNT1 and GALNT2 mix led to glycosylation of the GALNT_L sites. This supports the previous observations of the STR-MUC1 peptide experiments that these sites are primarily targeted by GALNT4.

Two further separate glycosylation experiments with C1GALT1 and ST6GALNAC1 were performed on the fully glycosylated peptide, from (GALNT1, GALNT2, GALNT4), using sub-equimolar donor concentrations relative to the available glycosylation sites. This was designed to reveal site preference rather than achieve full glycosylation. The resulting products were digested with extended IMPa cleavage to generate fragments for site mapping. The IMPa cleaves N-terminal to O-GalNAc residues except when preceded by aspartic acid, which limited the precise assignment of T8, S9, and T13. Despite this limitation, LC–MS analysis indicated sialylation at one site in the

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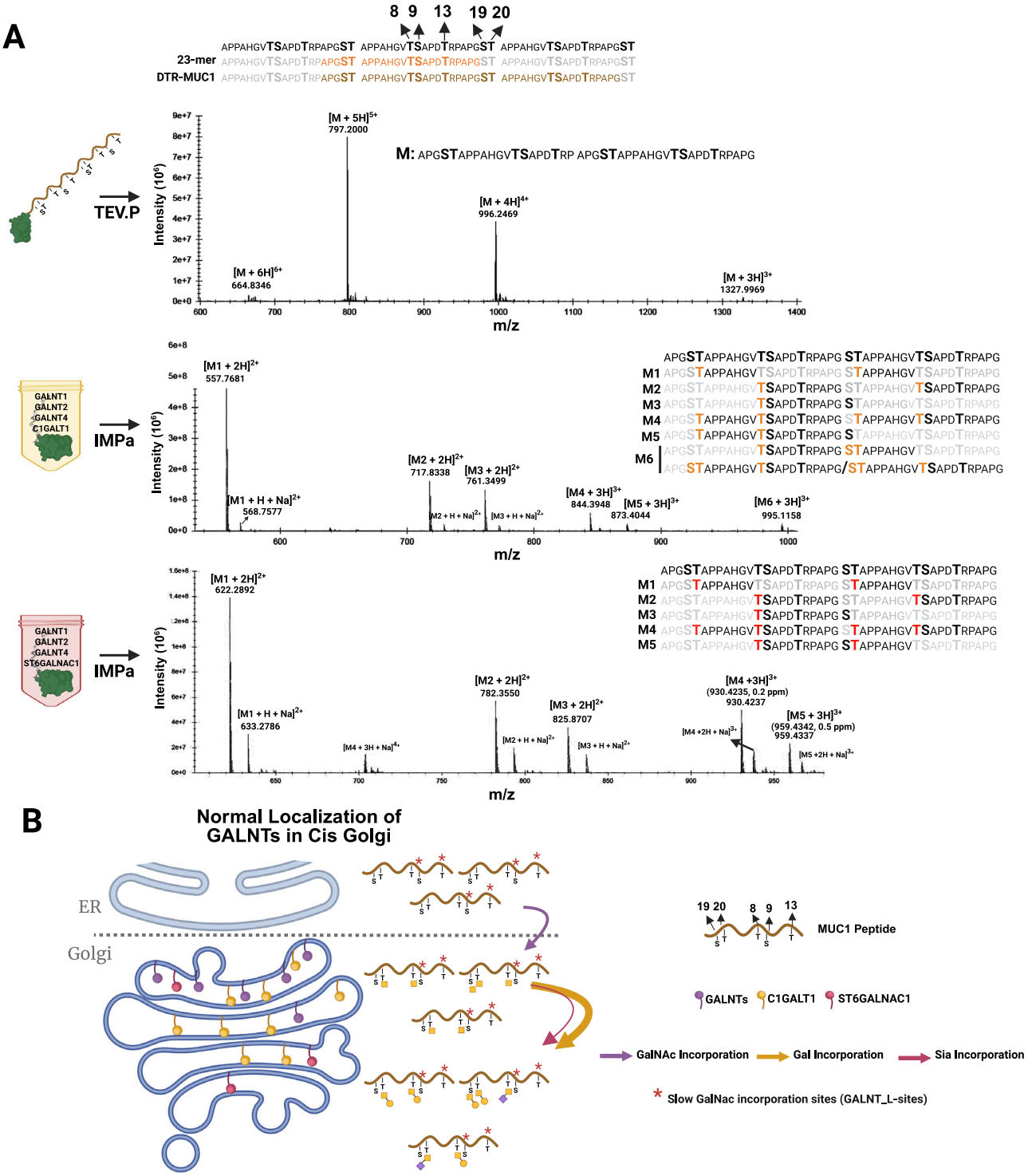
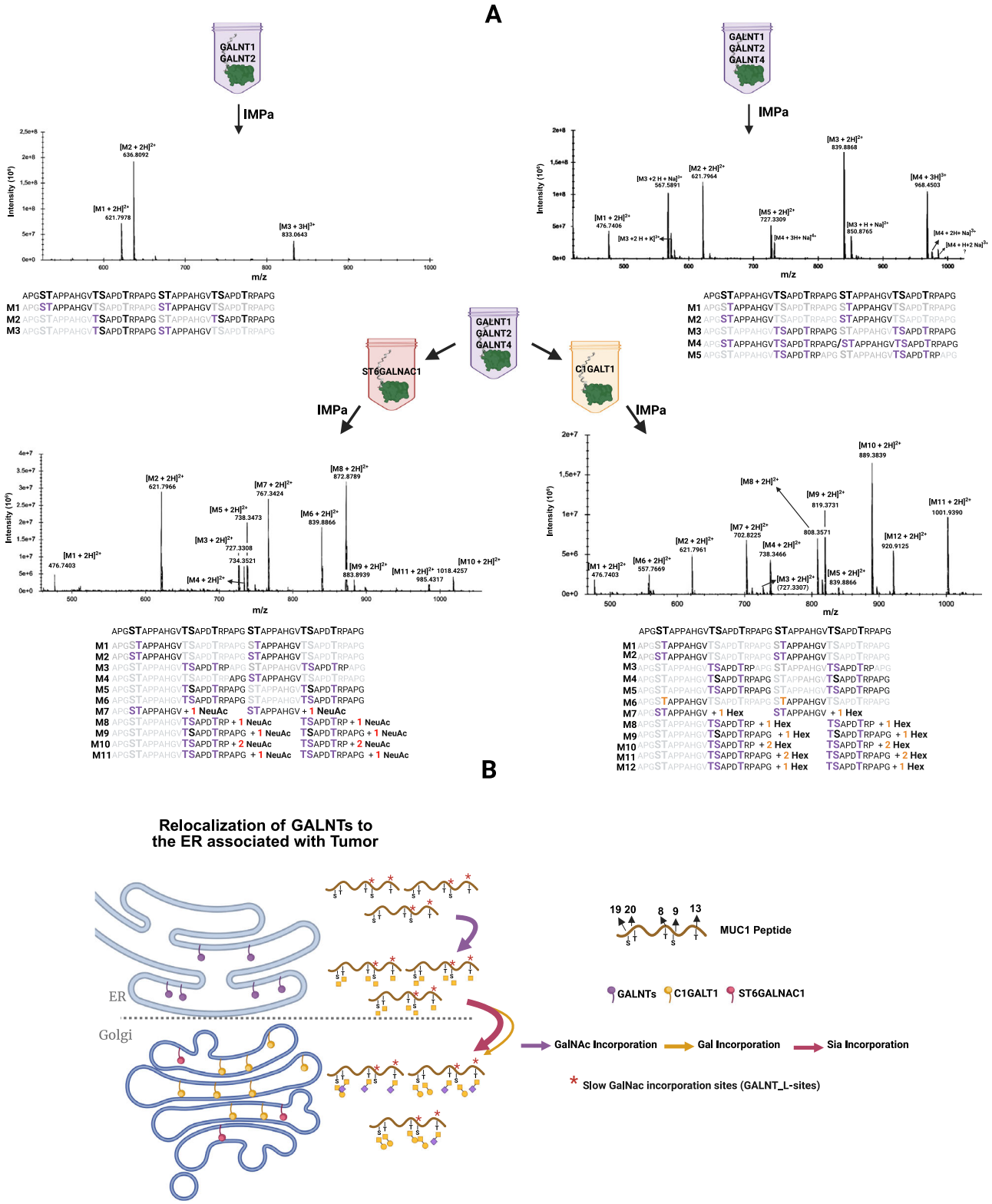


Fig. 6 | A proposed model of the effect of the colocalization of GALNTs with C1GALT1 and St6GALNAC1 on site occupancy and sialylation of the Tn antigen. The alignment of the Double Tandem Repeat (DTR)-MUC1 peptide with the Single Tandem Repeat (STR)-MUC1 23-mer (23-mer) peptide is shown in panel A. The DTR-MUC1 was expressed in fusion form using the same tag elements employed for the 23- and 27-mer peptides. **A** MS1 spectra obtained from peptide-related LC peaks are shown alongside the corresponding sample preparation schematic and the digestion enzyme used to release the peptide from the fusion protein (TEV protease or IMPa). Details of the LC–MS analyses are provided in the Supplementary Information (Supplementary Figs. 14–16). Sites bearing the T antigen are highlighted in orange, whereas sites carrying the sialyl-6-T antigen are indicated in red. Note: IMPa digestion was performed under mild conditions (see Supplementary Information figure captions); therefore, partially digested fragments are also observed. Because the peptide contains identical repeats, ambiguous assignments are shown at equivalent sites on both N- and C-terminal repeats. Source data are provided as a Source Data file. **B** Schematic representation of the proposed effect of GALNTs colocalization with C1GALT1 and ST6GALNAC1 on glycosylation site occupancy. The model shows that colocalization limits the elongation and sialylation of the Tn antigen to GALNT_D sites. In contrast, glycosylation is reduced at the S19 site or absent at GALNT_L sites. Created in BioRender. Naidoo, K. (2026) https://BioRender.com/qban85z.

https://doi.org/10.1038/s41467-026-72151-y



$S_{19}T_{20}$APPAHGV fragment and one or two sites in the T_8S_9APDT$_{13}$RPAPG fragment. While HCD fragmentation could not retain sialylated diagnostic ions, prior studies demonstrated that ST6GALNAC1 cannot sialylate Tn sites on serine residues[33]. Collectively, this supports that observed sialylation occurs at T20, T8, and T13. This aligns with the protein-protein docking findings in our study, which show that GalNAc residues at T13, T8, and T20 are preferentially positioned in the catalytic domain for sialylation by ST6GALNAC1 compared to S9 and S19.

Given these findings, the apparent K_m values for ST6GALNAC1 (Fig. 4A, table) were recalculated by correcting the substrate concentration to only account for sites participating in the reaction, excluding serine residues. The corrected Km values were 0.041 ± 0.019 and 0.068 ± 0.033 for GalNAc3-23mer and GalNAc5-23mer substrates, respectively, indicating no statistically significant difference, in contrast to the initial calculation in which all potential sites were assumed to participate in catalysis. The corresponding catalytic efficiencies calculated from the corrected Km values were 13.4×10^2 M-1. S-1 for

Fig. 7 | A proposed model of the effect of the ER localization of GALNTs on site occupancy and the subsequent glycosylation of the Tn antigen. A MS1 spectra obtained from integrated peptide-related LC peaks is shown beneath the corresponding sample preparation schematic and the digestion enzyme used to release the peptide from the fusion protein (IMPa). Details of the LC–MS analyses are provided in the Supplementary Information (Supplementary Figs. 17–20). Sites bearing the Tn antigen are highlighted in purple, whereas the numbers of sites carrying the T antigen or the sialyl-6-T antigen are indicated in orange and red, respectively. When a glycosylation site is confidently assigned, it is indicated directly in the sequence. Note: IMPa digestion of C1GALT1 and ST6GALNAC1 reaction products was performed under extensive conditions (see Supplementary Information figure captions), which may have contributed to non-specific C-terminal cleavage observed in some peaks. Because the peptide contains identical repeats, ambiguous assignments are shown at equivalent sites on both N- and C-terminal repeats. Source data are provided as a Source Data file. **B** Schematic representation of the proposed effect of GALNT localization in the ER on glycosylation site occupancy. The model shows that ER localization enhances occupancy at all GALNT-D and GALNT_L sites. In support of the kinetic and computational analyses, the LC–MS data suggest that ST6GALNAC1 exhibits preferential activity toward the T13 site, followed by T20 and T8. Created in BioRender. Naidoo, K. (2026) https://BioRender.com/d7iikxz.

GalNAc3-23mer and 14.5 × 10^2 M-1. S-1 for GalNAc5-23mer. While kinetic analyses suggest that the observed preference for GALNT_L sites is not driven by differences in apparent substrate affinity, computational analyses of individual sites indicate that the PDTR region containing T13 preferentially binds the catalytic domain and exhibits the lowest transition-state energy barrier among all sites examined. In contrast, both computational and LC-MS data indicate that S9 does not significantly contribute to specificity. Together, these findings provide a mechanistic basis for the enhanced apparent Vmax and kcat observed when GALNT_L sites are available, with this effect being primarily driven by T13.

In the C1GALT1 reaction, LC-MS analysis indicated preferential addition of galactose at one site in the S19T20APPAHGV fragment and one or two sites in the T8S9APDT13RPAPG fragment. However, detailed site specificity can be investigated by combining the same computational approaches used for ST6GALNAC1 in this work with site-resolving LC-MS/MS fragmentation methods (e.g., ETD or ECD); given the particular clinical relevance of the ST6GALNAC1 product sialyl-Tn (sTn) as a cancer-associated glycoepitope, ST6GALNAC1 was examined here as a proof of concept, while comprehensive analysis of C1GALT1 is left for future work.

Discussion

The in vitro reconstruction of the MUC1 GalNAc-*O*-glycosylation pathway delivered insights and details of (a) the competitive specificity of each enzyme to the substrate's MUC1 locale, (b) the possible glycosylated combinations of MUC1 (glycoforms) produced by each enzyme as well as the predominance of the glycoforms, (c) the effect of the sequence of the GTs in the glycosylation procession and (d) the effect of GT compartmentation (co-localization) on the glycosylation products profile. To illustrate the utility of the approach, the reported effect of GALNTs relocating to the ER on site occupancy was confirmed. However, insight into the nature of the GALNT relocation effect showed that two mechanisms are at the root of this effect. Firstly, the isolation of the GALNT in the ER extends the exposure to the substrate and so the time to react, which leads to complete glycosylation and the saturation of the MUC1 glycosylation target sites through lectin-dependent mechanisms. Secondly, further glycosylation of the GALNT_D-sites by C1GALT1 and ST6GALNAC1 leads to the inhibition of the lectin-dependent mechanism of GALNTs, so preventing complete saturating MUC1 GALNT_L-sites with a primary GalNAc.

The in vitro enzyme specificity and competition at each step of the synthesis revealed that B3GNT6 specificity to MUC1 Tn sites is negligible compared with C1GALT1 and ST6GALNAC1. This finding corresponds with previous summations that core 3 and core 4 structures are less predominant in MUC1. Previously, a comparison between normal epithelial breast cell lines and breast cancer cell lines recorded the absence of core 3 and core 4 structures from both types of cells[44]. Of greater note, the hypothesis that ST6GALNAC1 and ST6GALNAC2 may have an equal propensity to Tn and T antigens was invalidated by the observation that the relative specificity of ST6GALNAC1 toward Tn and T antigens is significantly greater than that of ST6GALNAC2. The conclusion is that ST6GALNAC1 and not ST6GALNAC2 is responsible for the α-2 sialylation of these two antigens. The reactivity of T antigen catalyzed by ST3GAL1 compared with ST6GALNAC1 showed that the former is more reactive, suggesting that the route to ST via sialyl-3-T is more likely than via sialyl-6-T.

A comparison of C1GALT1 and ST6GALNAC1 activity in completely vs. partially glycosylated MUC1 Tn antigen reactions revealed that C1GALT1 preferentially catalyzes reactions that incorporate galactose at GALNT_D sites (T8, S19, and T20), whereas ST6GALNAC1 is more active in sialylating the fully glycosylated GALNT_L sites (S9 and T13). Combining the results of the LC–MS analysis of ST6GALNAC1 sialylation of MUC1, with computational simulations of the sialylation reaction, establishes that T13 is the primary catalytically favorable site within the GALNT_L region, rather than S9. Revisiting the data from the chemoenzymatic synthesis approach undertaken by Yoshimura et al.[33], it is apparent that ST6GALNAC1 displayed significant specificity in catalyzing the reaction at threonine T13 compared with T8 and T20. In the same study, neither S9 nor S19 was sialylated by ST6GALNAC1. The reaction dynamics simulations provide the molecular rationale for this observation.

Assembling all the experimental, particularly the one-pot DTR-MUC1 biosynthesis, and the reaction dynamics simulations results makes possible the construction of mechanistic models that explain the differences between normal and tumor-associated MUC1 glycosylation (Figs. 6B and 7B). These models reveal that in normal epithelial cells, the co-localization of GALNTs with C1GALT1 and ST6GALNAC1 in the cis-Golgi leads to competition between these enzymes, which limits the formation of GALNT_L-sites. Consequently, the GALNT_D-site glycosylation is the primary product. In contrast, in tumor cells, the relocation of GALNTs to the ER, away from the competitive cis-Golgi glycosyltransferases, allows slower GalNAc catalysis to generate GALNT_L-sites in addition to GALNT_D-sites. The GALNT location leads to either normal MUC1 (GALNT_D-sites only) or tumor-associated MUC1 (TA-MUC1, with both GALNT_D- and GALNT_L-sites). These then serve as substrates for C1GALT1 and ST6GALNAC1. C1GALT1 preferentially extends GALNT_D-sites, enriching core 1 glycans typical of normal epithelial MUC1, whereas the presence of GALNT_L-sites in TA-MUC1, particularly T13, presents a favored substrate for ST6GALNAC1-mediated sialylation, leading to high-density synthesis of the tumor-associated sTn antigen.

Methods

Expression of MUC1 peptide fusion protein and tag removal

Gene sequences encoding MUC1 peptide fusion proteins were synthesized and inserted into the pET-21b(+) *E. coli* expression vector by BIOMATIK (Ontario, Canada). The sequence of the synthesized fusion protein is provided in the supplementary information (Supplementary Fig. 1). Insertion was performed at the 5′ HindIII and 3′ XhoI restriction sites. Transformation of *E. coli* BL21 (Sigma-Aldrich, Cat. no. CMC0014) was carried out using the heat shock protocol and grown on Luria Broth (LB) agar plates supplemented with ampicillin to a final concentration of 50 μg/ml at 37 °C. Expression was carried out in Terrific Broth (TB) medium overnight at 16 °C with shaking at 150 rpm. Glucose was added to a concentration of 2% wt/vol at all stages of expression.

Expression was induced by isopropyl β-d-thiogalactoside (IPTG, Sigma-Aldrich, Cat. No. 16758) to a final concentration of 1 mM at OD600 of 0.4–0.6.

The cell pellets from *E. coli* expression were lysed by incubating at 4 °C for 4 h in a binding buffer consisting of 20 mM Tris-HCl, 5 mM imidazole-HCl, 500 mM NaCl, 0.05% (w/v) sodium azide, and 10% (v/v) glycerol at pH 7.9. This buffer was supplemented with a protease inhibitor tablet (complete, Sigma Aldrich, Cat. no. 11873580001) and 20 mg of lysozyme per 10 ml of buffer. The cell lysate was then clarified through centrifugation at 48,000 × *g* for 30 min followed by filtration using a 0.45 μM sterile filter. Immobilized Metal Affinity Chromatography (IMAC) was conducted on a protein liquid chromatography (FPLC) system ÄKTA Start utilizing 1 ml HiTrap Chelating High-Performance columns (Cytiva, Cat. no. 17-0408-01). Proteins were eluted with a gradient ranging from 5 to 500 mM imidazole-HCl over 15 min at a flow rate of 1 ml/min. The eluted fractions were collected, and SDS-PAGE was used to verify protein purity and size. The purified enzymes were quantified with the Bradford protein assay kit (ThermoFisher, A55866) and stored at −80 °C in a freezing buffer comprising 20 mM Tris-HCl, 150 mM NaCl, and 10% (v/v) glycerol at pH 7.6.

TEV protease was expressed from the expression plasmid pRK793 (Addgene plasmid #8827; http://n2t.net/addgene:8827; RRID: Addgene_8827), in *E. coli* BL21(DE3)-RIL cells[14]. Tag removal was performed overnight at 4 °C by incubating the TEV protease with the fusion proteins at an optimized ratio (Supportive Information, Supplementary Fig. 2) in a buffer of 50 mM Tris-HCl, 0.5 mM EDTA, 1 mM DTT, pH 8.0. IMPa was supplied from Struenodo (Cape Town, South Africa). Cleavage of the fusion tag and digestion of the fusion protein were carried out using IMPa under either mild or extensive digestion conditions, depending on the experiment. Mild digestion was performed for 3 h using 1 unit of IMPa per 15 μM fusion protein in a 20 μl reaction, while extensive digestion was conducted under the same enzyme-to-substrate ratio for an extended incubation period. One unit of IMPa is defined as the amount of enzyme required to cleave 300 pmol of GFP-tagged MUC1 glycopeptide in a total reaction volume of 20 μl (substrate concentration 15 μM) after incubation for 2 h at 37 °C. Details of specific digestion conditions for each sample are explained in the Supportive Information.

Expression of the 23mer, the 27mer, and the DTR-MUC1 peptides was carried out in *E. coli* BL21 (DE3). A single-step purification using IMAC yielded 24-60 mg of pure fusion protein from 250 ml cell culture. Tag cleavage using TEV protease was performed, and SDS-PAGE gel confirmed complete cleavage of the naked peptide and the peptide displaying glycosylation on various sites (Supplementary Fig. 2).

Expression of glycosyltransferases

HEK293F cells (R79007, Thermo Fisher) were a gift from E. Sturrock (University of Cape Town, South Africa). All expression plasmids for GTs used in this work were purchased from the plasmid repository DNASU: HsCD00522282 for GALNT1, HsCD00413124 for GALNT2, HsCD00413161 for GALNT4, HsCD00413129 for GALNT7, HsCD00413109 for ST6GALNAC2, HsCD00413042 for C1GALT1C1 and HsCD00413169 for ST3GAL1. pGEn2 vectors of C1GALT1, ST6GALNAC1 and B3GNT6 were obtained directly from Kelley Moremen (University of Georgia, The USA). GTs expressed from pGEn2 vectors are *N*-terminally tagged with signaling peptide-8X His-Avi tag-superfolder GFP and TEV protease recognition sites. HEK293F cells (> 95% viability) were diluted to 1×10^6 cells/ml in 100 ml Freestyle media and incubated for 24 h. Cells were then collected and resuspended in 50 ml Freestyle media to $2.5–3 \times 10^6$ cells/ml, followed by a 30 min recovery period. Plasmid DNA (150–300 μg) was added dropwise, followed 5 min later by 450 μl of 0.5 mg/ml PEI (prepared by diluting 1 mg/ml stock 1:1 with media). Cultures were incubated for 24 h, then supplemented with 50 ml Freestyle media containing 4.4 mM valproic acid to achieve a final volume of 100 ml and 2.2 mM valproic acid. Expression was monitored daily by measuring fluorescence in the media; cells were harvested 3–6 days post-transfection once fluorescence plateaued, and viability dropped below ~50%. Cells were pelleted, and the media were filtered (0.45 μm) and directly loaded onto an IMAC column for purification of secreted glycosyltransferases. To remove the tag and obtain the enzymes in their native sequence, purified (tagged) proteins were incubated overnight at 4 °C with TEV protease at a ratio of 1:5 (TEV protease: fusion protein) in TEV protease buffer (50 mM Tris-HCl, 0.5 mM EDTA, 1 mM DTT, pH 8.0). The mixture was then processed by an IMAC to collect the untagged enzymes from the flow-through fractions. Enzymes were stored at −80 °C in freezing buffer (20 mM Tris-HCl, 150 mM NaCl and 10% (v/v) Glycerol, pH 7.6).

Enzyme assay

The enzymatic reactions were monitored using the previously developed UGC assay[14], the components of the assay are: l-Lactic dehydrogenase (LDH, L2500), PK (P1506), donors such as CMP-sialic acid (CMP-Neu5Ac, C1006), phosphoenolpyruvic acid monopotassium salt (PEP-K, 860077), adenosine 5′-triphosphate disodium salt trihydrate (ATP, 10519979001), β-Nicotinamide adenine dinucleotide, reduced disodium salt hydrate (NADH, 10128023001), bovine serum albumin (BSA, A3059), *N*-(2-hydroxyethyl) piperazine-N′-(2-ethanesulfonic acid), 4-(2-hydroxyethyl) piperazine-1-ethanesulfonic acid (HEPES, H3375), all purchased from Sigma Aldrich. The 27-mer peptide used as a standard naked peptide in Fig. 2B was synthesized at GL Biochem, Shanghai. Nucleoside diphosphate kinase (NDK) and cytidylate kinase (CMK) were expressed in-house as described in the original protocol. Peptides used as fusion proteins were quantified using the Bradford assay. The concentration of the GT enzymes was standardized in all the kinetics and activity assays at 100 nM. Donor concentration was standardized at 2 mM. The induction of reaction and preparation of reaction mixtures were performed following the standard protocol[14]. Briefly, Master mixtures were prepared in CMK-NDK-PK-LDH format for sialyltransferase enzymes and in NDK-PK-LDH format for the other enzymes. In addition to the coupling enzymes, master mixtures included: reaction buffer, BSA, NADH, phosphoenolpyruvate, ATP, donor, and the constant reaction components (enzyme or acceptor). The variable reaction component was distributed to the wells in triplicate, and the induction was carried out by adding the master mixture to the wells simultaneously. Corresponding controls, including reactions without the variable component, were also included. Fluorescence signals were recorded in real time, and initial rates were calculated from the slope of progress curves within the initial linear range and normalized to the corresponding blank controls. The obtained rates (in RFU/min) were converted to molarity per minute (In Fig. 4A) using the slope of the NADH standard curve ([NADH] vs. RFU). The resulting rates were then normalized by dividing by a factor of 1 for reactions with UDP-based donors and by a factor of 2 for reactions with CMP-based donors.

Preparative glycosylation

The standard glycosylation of the peptides in their fusion form was performed in a one-pot reaction mixture containing the peptide fusion protein at a concentration of 100 μM, donors at 2 mM and GT enzymes at 560 nM. The buffer used was 50 mM HEPES-NaOH buffer, 50 mM KCl, 10 mM $MnCl_2$, 10 mM $MgCl_2$, and 0.02 vol/vol Triton X-100 pH 7.4. A total of 1 ml reaction mixtures were prepared and incubated for 8 h at 37 °C. For Subsequent glycosylation of the DTR-MUC1 by C1GALT1 or ST6GALNAC1 after GALNTs, ST6GALNAC or C1GALT1 were added to the GALNTs reaction mixture directly (without an intermediate purification step). The reaction was performed under sub-equimolar donor concentrations relative to the available glycosylation sites (around 0.4 donor molecules per available site). After completion of the reaction, the glycosylated peptides in their fusion form were purified in a one-step IMAC. The eluted fusion proteins were then

subjected to buffer exchange to either freezing buffer (20 mM Tris-HCl, 150 mM NaCl and 10% (v/v) Glycerol, pH 7.6) when a subsequent glycosylation or IMPa digestion reactions are performed or to TEV protease cleavage buffer (50 mM Tris-HCl, 0.5 mM EDTA, 1 mM DTT, pH 8.0) when LC-MS is performed.

LC-MS analysis

LC–MS analysis of peptide and glycopeptide samples was performed using either a Q-Exactive Orbitrap Mass Spectrometer (Thermo Fisher Scientific) or a SYNAPT G2 Mass Spectrometry System (Waters Corporation). Data acquisition for the Q Exactive system was carried out using Xcalibur (v4.1.31.9) and Orbitrap MS v2.9, whereas the SYNAPT G2 system was operated using MassLynx (v4.1) in positive electrospray ionization mode with MSe continuum acquisition.

Peptide samples were prepared in a solution containing 0.1% (v/v) formic acid (FA) and 2% (v/v) acetonitrile (ACN). The final peptide concentrations were approximately 10 nM, with injection volumes corresponding to ~50 fmol and ~190 fmol per analysis, for Q-Exactive and SYNAPT G2, respectively, as detailed in the Supplementary Methods. For Q Exactive analyses, samples were trapped on a PepMap100 C18 trap column and separated on a ReproSil-Pur 120 C18-AQ analytical column (Dr. Maisch, r119.aq.n150.075, 75 µm × 15 cm × 1.9 µm). For SYNAPT G2 analyses, samples were separated on an XBridge Peptide BEH C18 XP column (Waters, 130 Å, 2.5 µm, 4.6 × 150 mm). Mass spectrometric data were acquired in positive electrospray ionization mode. Instrument parameters, chromatographic gradients, column specifications, and sample injection volumes are detailed in the Supplementary Methods.

Each peptide and glycopeptide sample was analyzed once ($n = 1$). No technical or biological replicates were performed, as samples were obtained from a single in vitro synthesis batch. Solvent-only blanks were injected between samples as negative controls. Theoretical m/z ratios and isotope distributions were predicted using the MS-Product (v6.6.6) and MS-Isotope tools from the Protein Prospector software suite (Baker and Clauser, http://prospector.ucsf.edu). Mass accuracy was determined by calculating the ppm difference between experimental and theoretical m/z values. Although the one-pot synthesis and single-step purification efficiently yield relatively pure glycopeptides, these preparations are classified as partially purified peptide mixtures due to residual buffer components and enzymes. Consequently, only MS1 peaks corresponding to the target peptides are presented. The complete, unfiltered raw data are available in the MassIVE database **MSV000099326** [https://doi.org/10.25345/C51V5BS8N].

Statistics and regression

All the data points were repeated three times. Shapiro–Wilk normality tests were performed to confirm the normal distribution of the data. Normality tests and Michaelis-Menten regression were performed using GraphPad Prism 8 software[14]. The data are presented as (mean value) ± (standard deviation). Kinetic parameters were determined by fitting data to the Michaelis-Menten equation using non-linear regression (GraphPad Prism). Initial 95% confidence intervals were calculated via the profile likelihood method; for reporting purposes, these were converted to standard error (SE) using a symmetrical approximation. Values in the table in Fig. 4A represent the best-fit parameter ± SE. Statistical comparisons between groups were performed using a two-tailed, two-sample equal variance Student's t-test; only P-values < 0.01 are displayed on the graphs, and comparisons not labeled were not statistically significant.

Computational methods

The ST6GALNAC1 structure was obtained from the SWISS-MODEL[45] database and used to prepare the Michaelis complex. This model was selected owing to its superior structural fidelity and similarity to the available crystal structure of the ST6GALNAC2 (PDB ID: 6APJ) compared with other models (Supplementary Figs. 21 and 22). The protein sequence of the CMP-binding position was determined by structural alignment with the crystal structure of ST6GALNAC2 (PDB ID: 6APJ), which was subsequently modified to CMP-Neu5Ac. The binding position of the MUC1 peptide was determined using the HADDOCK webserver [https://wenmr.science.uu.nl/haddock2.4/] to perform protein-protein docking. Details of the docking procedure are given in the Molecular docking section in the Supplementary Information.

GalNAc was included in the serine and threonine residues, and the resulting glycopeptide structures were energy-minimized. Following this, the Michaelis complex for each system was modeled with the CHARMM36 force field[46]. All systems were solvated in TIP3P water molecules in a cubic water box with a 12 Å buffer from the edge of the protein. The system was neutralized, and NaCl ions were added to a concentration of 0.15 M. Initial energy minimization was performed for 1,000 steps using the ABNR method, followed by a 1000 step minimization using the SD method. Thermal equilibration was performed for 100 ns with an NPT ensemble, and a 100 ns classical molecular dynamics simulation was performed in an NVT ensemble using NAMD3 (version 3.0.1)[47]. Particle-mesh Ewald (PME) summation was used to calculate electrostatic interactions. The Verlet cutoff scheme was applied to van der Waals (vdW) interactions with a cutoff distance of 12 Å and a switch distance of 10 Å. The SHAKE algorithm is used in all simulations to constrain hydrogen bonds. Simulations were carried out with a time step of 2 fs. All simulations were performed at a temperature of 300 K and under a pressure of 1 bar.

The reaction dynamics simulations were modeled using hybrid quantum mechanical/molecular mechanical methods (QM/MM) in CHARMM 37b2[48] compiled with SLASH[34] and FEARCF[49]. The QM/MM simulations were performed with identical simulation conditions to provide qualitative, comparative mechanistic insight into site-specific sialylation rather than to derive absolute kinetic or energetic parameters. The QM region comprised His567 (catalytic base) sidechain atoms, His552 (stabilizing the CMP phosphate group), the Neu5Ac moiety, the CMP phosphate group and the GalNAc acceptor. The nonpolar C-C bonds were treated with hydrogen link atoms, and the polar C-O bonds were treated with the Simple Link Atom Saccharide Hybrid (SLASH)[34] method.

Free energy simulations were conducted along two reaction coordinates to monitor bond formation and breaking using the Free Energies of Adaptive Reaction Coordinate Forces (FEARCF)[35,36] method. These were expressed as a linear combination of bond forming (C2-O6) and bond breaking (C2-O2) primary reaction coordinates (Fig. 5C). Histograms for each reaction coordinate were generated using 110 bins spanning a sampling range of 0.5–6 Å. Twelve FEARCF iterations were performed, each comprising 120 simulations of 30ps duration that were preceded by a 2 ps equilibration run, culminating in production runs each totaling 43.2 ns simulation time. Minimum energy pathways (MEPs) were determined from free energy surfaces generated from the entirety of the 43.2 ns production run.

All QM/MM and FEARCF simulations were performed in a 33 Å water sphere using stochastic boundaries that cover the full enzyme. A buffer shell (28–33 Å from the system origin) was applied. The reaction region was modeled with the mio1-1 DFTB3 parameter set, which includes corrections for hydrogen bonding and dispersion interactions with the dispersion corrections. SCC-DFTB[50] has been shown to be a suitable method for studying reaction mechanisms involving carbohydrate ring puckering, as previously demonstrated by Barnett et al.[51]. Forces between partial charges beyond 12 Å were zeroed using an atom-wise force-shifting function. Dynamics in the system were modeled using Langevin dynamics with a 2 fs time step.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

The LC-MS proteomics data generated in this study have been deposited in the MassIVE repository under accession number **MSV000099326** [https://doi.org/10.25345/C51V5BS8N]. Structural data of the ST6GALNAC2 enzyme used in this study is available in the RCSB PDB 6APJ. The QM/MM trajectories and free energy surfaces are proprietary and, due to impending patent filing, are not publicly available, but can be accessed from the corresponding author upon request and signing of a confidentiality agreement. The processed free energy data derived from these trajectories and surfaces are available in Fig. 5C and in the Source Data file. Initial coordinates used for reaction dynamic simulations and reaction visualization are available on GitHub [https://github.com/scientificomputing/NCOMMS-25-06286-T-MUC1-in-vitro-Synthesis-/tree/main] and Zenodo [https://doi.org/10.5281/zenodo.18903169]. Unless otherwise stated, all data supporting the results of this study can be found in the article, supplementary, and source data files. Source data are provided with this paper.

Code availability

The original version of the FEARCF code is embedded in the CHARMM package, which is accessible through an academic license. The updated FEARCF library and SLASH algorithms have been published and are available in papers cited here. The executables utilized for reaction dynamics simulations are available under restricted access as this proprietary code; access can be obtained by contacting the corresponding author. Classical molecular dynamics was run using NAMD3 (version 3.0.1) [https://www.ks.uiuc.edu/Research/namd/alpha/3.0alpha/] and QM/MM simulations were performed using CHARMM 37b2 [https://academiccharmm.org/], both of which are available from the respective developers under academic license.

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Author contributions

K.J.N. funded, supervised the study and developed the algorithms and code for QM/MM reaction dynamics. K.J.N. and A.N. conceived of the project of simulating tumor vs. normal O-glycosylation of MUC1 peptides. A.N. refined the strategy, developed and performed the experiments and analyzed the associated data. T.S. developed the QM/MM modeling protocol for ST6Galnac1 catalysis. K.D. performed the QM/MM simulations and analyzed the simulations and docking with guidance from T.S. K.J.N. designed the manuscript draft. A.N prepared the manuscript, and K.J.N. reviewed and edited the manuscript. All authors approved the final version of the manuscript.

Competing interests

K.J.N. declares that the structures identified through QM/MM trajectories in this work have potential commercial applications, including, amongst others, drug design and development. K.J.N intends to pursue intellectual property protection that may include this structural data; therefore, this has not been included in this publication as it is not central to the argument. K.J.N. declares that the reaction energetics provided are sufficient to support all experimental conclusions presented in this work. K.J.N. holds Struenodo shares. Struenodo provided experimental materials free of charge, as declared in the methods. All materials are commercially openly available. The remaining authors declare no competing interests.

Additional information

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Correspondence and requests for materials should be addressed to Kevin J. Naidoo.

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